METRIC

MIL-W-10842A
4 October 1982
SUPERSEDING
MIL-W-10842(CmlC)
17 January 1951

MILITARY SPECIFICATION

WAX, MICROCRYSTALLINE (METRIC)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of microcrystalline wax.

1.2 Classification. The microcrystalline wax shall be of the following types, as specified (see 6.2):

Type I - Hard wax
Type II - Soft wax
Type III - High melting point wax

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9160

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:28Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve Test

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Ammunition Lot Numbering
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D94 - Saponification Number of Petroleum Products
D127 - Drop Melting Point of Petroleum Wax Including Petrolatum
D445 - Kinematic Viscosity of Transparent and Opaque Liquids (And the Calculation of Dynamic Viscosity)
D482 - Ash From Petroleum Products
D1193 - Reagent Water
D1321 - Needle Penetration of Petroleum Waxes
D1500 - ASTM Color of Petroleum Products (ASTM Color Scale)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Source: https://assist.dla.mil -- Downloaded: 202610-17T14:28Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Chemical and physical characteristics. The microcrystalline wax shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Type I	Type II	Type III	Test paragraph
Saponification number, maximum	3	3	3	4.2.4.1
Melting point, °C:				4.2.4.2
Minimum	60	66	82	
Maximum	77	77	96	
Penetration, 25°C, maximum	35 (3.5 mm)	170 (17.0 mm)	35 (3.5 mm)	4.2.4.3
Color, maximum	8.0	Dark brown or lighter	0.5	4.2.4.4
Kinematic viscosity at 100°C, centistokes				4.2.4.5
Minimum	11.2	11.2	11.2	
Maximum	25.0	26.0	25.0	
Ash, percent by weight, maximum	0.03	0.03	0.03	4.2.4.6
Particle size, percent by weight passing:				4.2.4.7
No. 20 sieve, minimum	---	---	99	
No. 100 sieve, minimum	---	---	95	
No. 270 sieve, maximum	---	---	20	

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the microcrystalline wax of one type produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:28Z
Check the source to verify that this is the current version before use.

continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 500 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container damaged or leaking
(c) Marking incorrect, missing, or illegible

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Saponification number. Determine the saponification number of the specimen in accordance with ASTM D94.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:28Z
Check the source to verify that this is the current version before use.

4.2.4.2 Melting point. Determine the melting point of the specimen in accordance with ASTM D127.

4.2.4.3 Penetration. Determine the penetration of the specimen at 25°C in accordance with ASTM D1321.

4.2.4.4 Color. Determine the color of types I and III microcrystalline wax in accordance with ASTM D1500. Determine the color of type II microcrystalline wax by visual examination.

4.2.4.5 Viscosity. Determine the viscosity of microcrystalline wax in accordance with ASTM D445.

4.2.4.6 Ash. Determine the percent by weight ash of the specimen in accordance with ASTM D482.

4.2.4.7 Particle size. Using sieves conforming to RR-S-366, nest a No. 20 sieve, a No. 100 sieve, and a No. 270 sieve, in order of fineness, with the most coarse sieve on top. Place the assembly on a bottom pan. Weight to the nearest 0.01 g approximately 100 g of the specimen and place it on the first (most coarse) sieve. Cover the sieves and shake for 30 minutes by means of a mechanical shaker geared to produce 300 $\pm$ 15 gyrations and 150 $\pm$ 10 taps of the striker per minute. Weigh the material retained on each sieve and calculate the percent by weight material passing through each sieve.

5. PACKAGING

5.1 Industrial unit packing, packing, and marking. A specified quantity of microcrystalline wax (see 6.2) shall be unit packed, packed, and marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Types I and II microcrystalline wax are intended for use as a water-resistant coating. Type III microcrystalline wax is intended for use as a binder in the manufacture of munitions.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Type of microcrystalline wax required (see 1.2)
(c) Unit quantity of microcrystalline wax required (see 5.1)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding off method of ASTM E29.

Source: https://assist.dla.mil -- Downloaded: 202610-17T14:28Z
Check the source to verify that this is the current version before use.

Custodian:

Army - EA

Review activities:

DLA - PS, GS

Preparing activity:

Army - EA

Project No. 9160-A066

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:28Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armement Research and Development Command
ATTN: DRDAR-TSC-S
Aberdeen Proving Ground, MD 21010

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil - Downloaded: 2026-10-17T14:28Z
Check the source to verify that this is the current version before use.